Osmotic Shock Transformation of D. salina

To get our target genes expressed in Dunaliella salina, we used an osmotic shock transformation method to deliver the assembled plasmids into the cells.

This section includes:

1. Cell culture and preconditioning
2. Screening and identification of transformants

Protocol – Osmotic Shock Transformation of D. salina

Cell culture and preconditioning

D. salina FACHB-435 was grown statically in modified DM medium (with 1.5 M NaCl) at 26°C, under 50 μmol photons m^{-2} s^{-1} of light and a 16 h daily photoperiod. The cells were harvested once they reached logarithmic phase (about 10^5–10^6 cells/mL), at which point the OD680 reads roughly 1.0.

1. Take the log-phase culture and pellet the cells in a centrifuge.
2. Pour off the supernatant, resuspend the pellet in fresh PKS/DM medium, and wash the cells 3 times to strip off residual metabolites from the old medium.
3. Resuspend the washed cells in fresh medium and adjust the final cell density to 10^6 cells/mL.
4. Transfer this suspension into fresh medium containing 1 M NaCl and keep culturing for another 8 h. This step "pre-adapts" the cells to the osmotic shock that follows.
5. Take 1 mL of the preconditioned suspension, spin it down, and discard the supernatant.
6. Immediately resuspend the pellet in fresh DM medium containing 0.1 M NaCl (bringing the volume back to 1 mL).
7. Add the following to this 1 mL suspension, in order and precisely:
    a) Plasmid DNA: 5.46 μg of the vector carrying the target gene
    b) Surfactant: 15 μL of 0.1% Triton X-100 stock
8. Vortex the mixture vigorously for about 90 s.
9. As soon as the vortexing is done, transfer the mixture into fresh liquid medium.
10. Let the cells recover statically for 48 h at 26°C under a 16 h daily photoperiod. By then, over 90% of the surviving transformed cells can return to normal growth.

Screening and identification of transformants

Fluorescence microscopy: cells from the recovery culture were collected and fixed with 4% paraformaldehyde for 30 min, washed twice by resuspension in PBS, and then examined under a fluorescence microscope. Because the vector carries EGFP, successfully transformed cells show green fluorescence.

Molecular-level identification (PCR): genomic DNA was extracted from the transformed cells and amplified with gene-specific primers. A band of the expected size confirms that the exogenous gene has been stably integrated into the cell genome.

Selection steps:

1. DM solid medium with a salt concentration of 0.5 M and 2% agar
2. Antibiotic concentration screen: take the same amount of D. salina as used after the osmotic transformation culture, collect the algal pellet by low-speed centrifugation, and plate it on D. salina plates with different antibiotics to pin down the minimum lethal concentration

3. Plate the post-transformation algae onto plates with the corresponding antibiotic; after they have grown for a while, pick single colonies and scale them up — these are the transformants
4. Extract the genomic DNA again and verify the transformants by qPCR

Plant Genomic DNA Extraction

To check whether the target gene had actually integrated into the chromosomal genome of D. salina, we extracted genomic DNA from the algae.

This section includes:

1. DNA extraction
2. DNA quantification and purity check

Protocol –Plant Genomic DNA Extraction

Genomic DNA extraction

We used the Plant Genomic DNA Kit from TIANGEN.

Before starting, add anhydrous ethanol to buffer GD and wash buffer PW — the volumes are specified on the bottle labels.

1. Take about 100 mg of fresh D. salina algal pellet (or ~30 mg of dried tissue) and grind it thoroughly in liquid nitrogen.
2. Quickly transfer the ground powder into a centrifuge tube pre-loaded with 700 μL of buffer GP1 pre-warmed to 65°C (β-mercaptoethanol was added to the pre-warmed GP1 beforehand, to a final concentration of 0.1%). Mix right away by inverting, then incubate the tube in a 65°C water bath for 20 min, inverting the tube a few times along the way to keep the sample mixed.
3. Add 700 μL of chloroform, mix thoroughly, and centrifuge at 12,000 rpm (~13,400 × g) for 5 min. Note: for tissues rich in polyphenols or starch, do an equal-volume phenol:chloroform (1:1) extraction before this step.
4. Carefully transfer the upper aqueous phase from the previous step into a fresh centrifuge tube, add 700 μL of buffer GP2, and mix well.
5. Load the mixture onto adsorption column CB3, centrifuge at 12,000 rpm (~13,400 × g) for 30 s, and discard the flow-through. (The column holds about 700 μL, so it can be loaded and spun in batches.)
6. Add 500 μL of buffer GD to column CB3 (double-check that anhydrous ethanol has been added), centrifuge at 12,000 rpm (~13,400 × g) for 30 s, pour off the waste, and put the column back into the collection tube.
7. Add 600 μL of wash buffer PW to column CB3 (double-check that anhydrous ethanol has been added), centrifuge at 12,000 rpm (~13,400 × g) for 30 s, pour off the waste, and put the column back into the collection tube.
8. Repeat step 7.
9. Put column CB3 back into the collection tube and centrifuge at 12,000 rpm (~13,400 × g) for 2 min; pour off the waste. Then leave the column open at room temperature for a few minutes so that any residual wash buffer in the adsorption material dries off completely. Note: the point of this step is to remove the residual wash buffer from the column — leftover ethanol will interfere with downstream enzyme reactions (restriction digestion, PCR, and the like).
10. Transfer column CB3 into a clean centrifuge tube, drip 50–200 μL of elution buffer TE onto the center of the adsorption membrane, leave it at room temperature for 2–5 min, then centrifuge at 12,000 rpm (~13,400 × g) for 2 min and collect the solution in the tube. Note: don't elute with less than 50 μL — small volumes hurt the recovery. The pH of the eluent matters a great deal: if ddH2O is used as the eluent, keep its pH between 7.0 and 8.5, since anything below 7.0 lowers

the elution efficiency. Store the DNA product at −20°C to keep it from degrading. For a higher genomic DNA yield, the collected solution can be reloaded onto column CB3, left for 2 min at room temperature, and spun again at 12,000 rpm (~13,400 × g) for 2 min.

DNA quantification and purity check

How large the recovered genomic DNA fragments come out depends on how long the sample was stored and how much shear it went through during handling. The recovered DNA can be checked on an agarose gel, with its concentration and purity read off a UV spectrophotometer.

DNA should give a strong absorption peak at OD260; an OD260 of 1 corresponds to roughly 50 μg/mL of double-stranded DNA, or 40 μg/mL of single-stranded DNA.

The OD260/OD280 ratio should sit between 1.7 and 1.9. If the DNA was eluted in ddH2O rather than elution buffer, the ratio tends to run low, because pH and ions shift the absorbance values — that doesn't by itself mean the DNA is impure.

Quantitative Real-Time PCR (qPCR)

To measure the copy number of the target gene in the D. salina genome, we ran qPCR.

This section includes:

1. Pre-experiment setup
2. qPCR reaction system
3. Loading
4. qPCR amplification program
5. Data processing

Protocol –Quantitative Real-Time PCR (qPCR)

Pre-experiment setup

1. Give the extracted D. salina genomic DNA a quick spin to bring everything down, mix briefly, and keep it on ice until use.
2. Thaw the primers and the qPCR master mix on ice, and work away from light throughout.
3. Set up the sample groups: a wild-type control group, several transgenic D. salina groups, and a blank (no-template) control.
4. Every sample, internal reference, and target gene gets 3 biological replicates plus 3 technical replicates.

qPCR reaction system

Component	Volume
2× SYBR Green qPCR Master Mix	10 μL
Forward primer (10 μM)	0.4 μL
Reverse primer (10 μM)	0.4 μL
Genomic DNA template	1 μL
Nuclease-free ultrapure water	8.2 μL
Total volume	20 μL

Loading

1. Set up the reactions on ice and out of the light — pure water and primers go in first, with the qPCR mix and the DNA template added last to keep bubbles down.
2. Mix gently by pipetting up and down 3–5 times; no vigorous shaking.
3. Quick-spin for 10 s to knock the liquid off the tube walls and clear any bubbles.
4. Load the 8-strip tubes into the block of the qPCR instrument, making sure they sit flat and stay out of the light.

qPCR amplification program

1. Initial denaturation: 95°C for 3 min
2. Cycling stage (40 cycles):
   95°C for 10 s (denaturation)
   58–60°C for 20 s (annealing, extension and fluorescence acquisition,fine-tuned to the primer Tm)
3. Melt curve stage:
   65°C → 95°C on a slow ramp, with fluorescence collected throughout
   used to judge product specificity — whether there is any non-specific amplification or primer dimers

Data processing

1. Export the raw Ct values, melt curves, and amplification curves once the run is over.
2. Look over the melt curves: a single, sharp, symmetric peak means specific amplification; data with extra peaks or primer dimers don't count as valid.
3. Pool the Ct values from each group, work out the means and standard deviations, and trim outliers.
4. Use an endogenous D. salina housekeeping gene as the internal reference to correct for DNA loading and extraction variation across samples.
5. Convert to relative genome copy numbers of the exogenous target gene with the $2^{-\Delta\Delta Ct}$ method.
6. Compare the signal between wild-type and transgenic lines to rank them by exogenous gene copy number, and pick out the stable, high-expression engineered strains.

Chromatographic Separation of the Products

To isolate and purify crocetin from the D. salina extract.

This section includes:

1. Sample pretreatment
2. Column packing and equilibration
3. Sample loading
4. Gradient elution and fraction collection
5. Concentration and storage

Protocol –Chromatographic Separation of the Products

Sample pretreatment

1. Centrifuge the D. salina fermentation broth and collect the algal pellet.
2. Break the cells open and extract the pigments with organic solvent to get a crude pigment extract.
3. Spin the crude extract at low speed to remove insoluble debris, then pass the supernatant through a 0.22 μm organic-solvent membrane filter to give a clear loading sample — this keeps impurities from clogging the column.

Column packing and equilibration

1. Pack the column by the wet method, loading the silica gel evenly into the column so that the bed stays level, with no bubbles and no channels.
2. Add the starting solvent and keep flushing the column, equilibrating the packing fully so that the environment inside the column is stable.
3. When the liquid level comes down just level with the bed, the column is ready for loading.

Sample loading

1. Add the pretreated crude pigment extract to the column slowly and evenly, letting it run down the column wall.
2. Keep the flow slow so that the sample adsorbs completely onto the surface of the silica bed, giving the first separation between the target product and the impurities.

Gradient elution and fraction collection

1. Elute stepwise with an organic solvent gradient, based on the polarity difference between astaxanthin and crocetin.
2. Collect the elution fractions in order of polarity:
   low-polarity impurities come off first
   the medium-polarity fraction is enriched in astaxanthin
   the high-polarity fraction is enriched in crocetin
3. Collect the eluate in fractions, label them in order, and hold a constant flow rate throughout.

Concentration and storage

1. Concentrate the product-containing fractions on a rotary evaporator to remove the organic solvent.
2. Redissolve the purified product in the corresponding chromatography solvent.
3. Store it cold and away from light, until the HPLC qualitative and quantitative analysis.